

1 Some Facts about Real-Analytic Functions

In this appendix we describe various elementary facts regarding (real-)analytic functions in a manner suitable for reference in the main text.

Let M be any analytic manifold of dimension l . (In this paper, “manifold” always means second-countable manifold.) Recall that an embedded submanifold Z of M , of dimension q , is a connected subset which, locally around each of its points and up to analytic diffeomorphisms, looks like a “slice” $\{(x_1, \dots, x_l) \mid x_{q+1} = \dots = x_l = 0\}$. When $q = l$, this is just an open set.

Assume now that the submanifold Z has positive codimension, that is, $q \leq l - 1$. From the definition, it follows that for some open subset $M_0 \subseteq M$, Z is a closed subset of M_0 (in the relative topology). Observe that Z is nowhere dense, that is, its closure has empty interior. (That is, if U is any open subset of M , then $Z \cap U$ cannot be dense in U : if U does not intersect M_0 then this is clear; otherwise $U \cap M_0$ is a nonempty open set and we may assume without loss of generality that $U \subseteq M_0$; then $U = (U \setminus Z) \cup (U \cap Z)$, so either $U \setminus Z$ is a nonempty open set, and we are again done, or $U \cap Z = U$, but in this latter case Z would contain an open set and hence could not have positive codimension.) Also, such a Z has measure zero.

For simplicity, if Z is a countable union of embedded analytic submanifolds of M of dimension $\leq q$, and q is the smallest such integer, we say that Z is a σ -analytic subset of M and call q the *dimension* of Z . (It is not hard to verify that the dimension is well-defined, in the sense that it doesn’t depend on the particular union of countably many submanifolds being used.) The σ -analytic subset Z of M will be said to be *analytically thin* if $\dim Z < \dim M$. Such a set has zero measure and is of the first category (a countable union of nowhere dense sets), as remarked above. Conversely, if a σ -analytic subset and it has measure zero then it must be analytically thin (otherwise it contains a submanifold of full dimension, that is, an open subset of M). We also use the following terminology: a subset Z of M is *finitely analytically thin* if it is a finite union of embedded analytic submanifolds of positive codimension.

In studying analytic sets and mappings, it is useful to introduce the notion of a semianalytic subset Z of an analytic manifold M . This is a set Z so that, for each $z \in M$, there is some neighborhood U of z so that $Z \cap U$ is in the Boolean algebra generated by a finite family of subsets of the form $\{f_j(x) > 0\}$, for some analytic functions $f_j : U \rightarrow \mathbb{R}$, $j = 1, \dots, J_z$. It is easy to see from the definition of embedded submanifold that if $Q \subseteq M$ is such a submanifold, then Q is a countable union of compact semianalytic subsets. It follows that every σ -analytic subset of M is a countable union of compact semianalytic subsets. Conversely, every semianalytic subset is a countable union of embedded submanifolds; see for instance Property 8(e) in [5]. In conclusion, being a σ -analytic subset is the same as being a countable union of semianalytic subsets (or of compact semianalytic subsets). We will use the following very special consequence of general stratification theorems (see for example Theorem 9.2 in [5], as well as [1]):

Fact 1.1 Let M and N be analytic manifolds and $f : M \rightarrow N$ an analytic mapping. Assume that Z is a compact semianalytic subset of M . Then there is a partition of N into a countable union of connected analytic embedded submanifolds Q_j , and a partition of Z into a countable union of connected analytic embedded submanifolds so that each such submanifold is diffeomorphic to $\mathbb{R}^{n_j} \times Q_j$, for integers $n_j \geq 0$ and suitable indices $j \in J_Z$, and on each such set the mapping f is (up to the same diffeomorphism) the projection $\mathbb{R}^{n_j} \times Q_j \rightarrow Q_j$.

Observe that it also follows from Fact 1.1 that the image $f(Z)$ is a σ -analytic subset of N (since it is a union of a subfamily of the Q_j ’s). Note that if Z is a σ -analytic subset of M ,

then the above discussion shows that $\dim Z = q$ if and only if Z can be written as a countable union of embedded submanifolds in such a way that the maximum of the dimensions of the submanifolds is q .

The next statements amount to saying that naive parameter counts are well-justified when dealing with analytic mappings.

Proposition 1.2 Assume that M , N , and M_i , $i = 1, \dots, k$, are analytic manifolds. Let $f : M \rightarrow N$ be an analytic mapping. Then:

1. If Z is a σ -analytic subset of M , then $f(Z)$ is a σ -analytic subset of N , and $\dim f(Z) \leq \dim Z$.

2. For all $Z \subseteq M$,

$$\dim Z \leq \dim f(Z) + \max_{y \in N} \left[\dim f^{-1}(y) \cap Z \right] .$$

3. If Z_i is analytically thin in M_i , for $i = 1, \dots, k$, then $Z = Z_1 \times \dots \times Z_k$ is analytically thin in $M_1 \times \dots \times M_k$ and

$$Z = Z_1 \times \dots \times Z_k \subseteq M_1 \times \dots \times M_k$$

satisfies $\dim Z = \sum_i \dim Z_i$.

Proof. In order to calculate the dimension of $f(Z)$, it is enough, by the above considerations, to do this when Z is a relatively compact semianalytic subset, and thus the dimension inequality follows by Fact 1.1.

To prove the statement about fibres $f^{-1}(y)$, we proceed as follows. Note that each such fibre is semianalytic, so its dimension is well-defined. Write Z as a countable union of compact semianalytic subsets Z_i ; then

$$\max_y \left[\dim f^{-1}(y) \cap Z \right] = \max_{y,i} \dim \left[f^{-1}(y) \cap Z_i \right] .$$

Fix any i , and apply Fact 1.1 with Z_i instead of Z . Thus $q = \max_y \dim [f^{-1}(y) \cap Z_i]$ is the largest of the n_j 's, while $\dim Z_i$ is at most $q + t$, $t =$ largest dimension of the Q_j 's, $j \in J_Z$, and $f(Z)$ has dimension t . This shows that

$$\dim Z_i \leq \dim f(Z) + \max_{y \in N} \left[\dim f^{-1}(y) \cap Z_i \right] \leq \dim f(Z) + \max_{y \in N} \left[\dim f^{-1}(y) \cap Z \right]$$

from which, since $\dim Z = \max_i \dim Z_i$, the conclusion follows.

Finally, to prove that $\dim Z_1 \times Z_2 = \dim Z_1 + \dim Z_2$, simply note that $Z_1 \times Z_2$ equals a union of the type $Z_1^j \times Z_2^k$, for countable coverings by submanifolds for each of Z_1 and Z_2 respectively, and dimensions add as they should for submanifolds. ■

2 Some Facts about Order-Minimality

Here we summarize certain recent facts from model theory used in order to prove the results given in the text.

Pick any positive integer l , and a cube $C = [-k, k]^l$ in $\mathbb{R}^l$. Assume that g is a real-valued function which is (real-)analytic in a neighborhood of C . By the θ -restriction of g to C we will mean the function $f : \mathbb{R}^l \rightarrow \mathbb{R}$ which equals 0 outside C and equals g on C . A *restricted analytic* (RA) *function* is any function obtained in this manner. Below we formally state what it means for a function to be (“exp-RA”) definable —informally, these are functions that can be defined in terms of a first-order logic sentence involving the standard propositional connectives, existential and universal quantification, algebraic operations, and symbols for the exponential function as well as all RA functions. Of course, $\tanh$ is definable, since $y = \tanh(x)$ if and only if $(y + 1)(1 + e^x) = 2$. Any RA function is in particular definable. The function $\arctan(x)$ is also definable, since $y = \arctan(x)$ if and only if $-\pi/2 < y < \pi/2$ and $\sin(y) = x \cos(y)$, where $\sin$ and $\cos$ denote the restrictions of $\sin$ and $\cos$ to $[-\pi/2, \pi/2]$. Compositions such as $\arctan(\exp(\exp(x)))$ are also allowed. (However, the function $\sin(x)$ is *not* definable.)

Formally, consider the structure

$$L = (\mathbb{R}, +, \cdot, <, 0, 1, \exp, \{f, f \in \text{RA}\}),$$

and the corresponding language for the real numbers with addition, multiplication, and order, as well as one function symbol for real exponentiation and one for each restricted analytic function. The set of (first order) formulas over L is the set of all well-formed logical expressions obtained by using propositional connectives, real numbers as constants, the operations of addition and multiplication, the relations $<$ and $=$, and $\exp$ and restricted analytic functions as functions; quantification is allowed over variables. (This is an example of a formula $\Phi(x, y)$ over L :

$$\forall z \left[e^{7z^2 e^y} - \pi x z \geq \arctan(e^x) \right].$$

We write $\Phi(x, y)$ to indicate the fact that the only free —i.e., non-quantified— variables in the formula are x and y .) Each such formula will be interpreted over the real numbers, that is, all variables are assumed to take real values. Thus all quantifiers are implicitly assumed to be over $\mathbb{R}$. Given a formula Φ with free variables $x_1, \dots, x_l$, we write $\mathcal{S}(\Phi)$ for the subset of $\mathbb{R}^l$ that it defines. A *definable* set is a set of the form $\mathcal{S}(\Phi)$, for some first order formula Φ over the language L . For instance, the above $\Phi(x, y)$ gives rise to:

$$\mathcal{S}(\Phi) = \left\{ (x, y) \in \mathbb{R}^2 \mid (\forall z \in \mathbb{R}) \left[e^{7z^2 e^y} - \pi x z \geq \arctan(e^x) \right] \right\}.$$

Similarly, the truth of a formula Φ with no free variables is defined as the truth of the statement obtained when quantifying over the reals. By abuse of notation, when giving such a formula, we will also allow other symbols, such as “ $-$ ” or “ $\geq$ ” which could be in turn defined on the basis of the above primitives, or even symbols for any set already known to be definable. A (*exp*-RA) *definable function* is a function $f : M \rightarrow N$ whose graph is a definable set in the above sense, and where N and M are definable subsets of two spaces $\mathbb{R}^{l_1}$ and $\mathbb{R}^{l_2}$ respectively.

When the exponential is left out, definable sets are precisely those called “finitely subanalytic” in [6]. Restricted analytic functions were introduced in [8]. (The definition in that reference is slightly different from the one we gave in the previous section: it assumes that the functions g have a convergent power series representation valid on all of the cube $C = [-k, k]^l$,

but a standard compactness argument shows that the two definitions are equivalent.) Gabrielov showed in [2] that the theory of real numbers with restricted analytic functions is model-complete, which means that every formula is equivalent to one that involves only existential quantification. (We do not give the precise statements here, as they are not needed for explaining the further material.) In a recent major development, Wilkie showed in [10, 11] that using exponentiation (but now leaving out the RA functions), model-completeness obtains as well. Finally, in [8] and [9], it was shown that the full theory (RA as well as exponentials) is model-complete, and hence order-minimal:

Fact 2.1 ([8], Theorem 6.9, and [9]) The theory of L is *order-minimal*, that is, for each formula Φ having just one free variable, $\mathcal{S}(\Phi)$ is a subset of $\mathbb{R}$ consisting of a finite union of intervals (possibly unbounded or just points).

The terminology reflects that such finite unions are the smallest Boolean algebra of subsets that can be defined using order. The forthcoming book [7] by van den Dries deals in detail with order-minimal theories. Sets definable (in any dimension) for order-minimal theories admit finite cell decompositions into topological submanifolds. In particular, this applies to parametric versions. Decompositions can be obtained which are uniform on parameters, and in particular the number of connected components is uniformly bounded. To be more precise, assume given a formula $\Phi(\Lambda, x)$, where Λ denotes a set of p variables and x denotes a set of q variables, for some integers p and q . For each fixed $\lambda \in \mathbb{R}^p$, we may consider the formula $\Phi_\lambda(x) = \Phi(\lambda, x)$ on the free variables x , and the respective definable set. It then follows from the general theory (see [4, 7]):

Fact 2.2 Given a formula Φ as above, there is an integer κ so that for all $\lambda \in \mathbb{R}^p$, the number of connected components of $\mathcal{S}(\Phi_\lambda)$ is at most κ .

When dealing as here with a language whose primitives stand for analytic functions, the cell decomposition results can be stated in a stronger fashion. By [8], Theorem 8.8, one knows that each definable subset is a finite union of what are called in that paper analytic cells, each of which is definable and definably-isomorphic to an Euclidean space. The definition of analytic cell in that paper implies that each such cell is an embedded analytic submanifold. Thus one also has the following result:

Fact 2.3 Let S be a definable subset of $\mathbb{R}^q$. Then, either S contains an open subset or it is finitely analytically thin.

Observe that a function such as $\sin(x)$ (seen as a function of $x \in \mathbb{R}$) is not definable, so there is no contradiction with the fact that its set of zeroes is not finitely analytically thin. (The zero set is of course analytically thin, consistent with the analyticity of $\sin(x)$.)

Note that a *finitely* analytically thin subset is nowhere dense (as it is a finite union of nowhere dense subsets). So this follows from Fact 2.3:

Corollary 2.4 If S is a definable subset of $\mathbb{R}^q$, then either it has nonempty interior or it is nowhere dense.

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